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MILITARY SPECIFICATION

ELECTRODES, WELDING, MINERAL COVERED, IRON-POWDER, LOW-HYDROGEN MEDIUM AND HIGH TENSILE STEEL, AS WELDED OR STRESS-RELIEVED WELD APPLICATION

This limited coordination military specification has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-E-22200/1. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used for acquisition purposes.

1. SCOPE

1.1 Scope. This specification covers iron-powder, low-hydrogen types of covered electrodes, for the fabrication and repair welding of low and medium carbon, high tensile and higher-strength low-alloy steels for as-welded and stress-relieved application.

1.2 Classification. Iron-powder, low-hydrogen electrodes shall be furnished in the types, classes, and sizes shown in table I, as specified (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE I. Types, classes, and sizes.

Type 1/ 2/ 3/	Class	Positions of welding	Sizes (diameter)
MIL-7018 MIL-8018-C3	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4
MIL-8018-C3SR	3	Flat	5/16
MIL-9018-M MIL-10018-M MIL-11018-M	1 2	All Horizontal fillets and flat	3/32, 1/8, 5/32 3/16, 7/32, 1/4

1/ Type classifications of electrodes will be determined according to the capability of the electrode to meet the tests specified herein, when used in the positions of welding specified in table I and in section 4. An electrode shall be approved under ONE type classification; and an electrode shall NOT be approved if it differs in type from that stated in the manufactuer's request for test. The position description indicates the maximum size qualified under the specification. However, reference should be made to the Qualified Products List (QPL) for the selection of electrodes for all positions of welding.

2/ The suffix RC added to the MIL type pertains to restriction of the copper content (see 3.4). All requirements, tests and acceptance criteria remain otherwise unchanged.

3/ Addition of the suffix SR to the basic MIL type designation indicates a special MIL type electrode formulated to meet the long time (50 hours) stress-relieved mechanical properties/requirements only.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-10430 - Welding Rods and Electrodes; Preparation for Delivery of.
MIL-E-22200 - Electrodes, Welding, Covered; General Specification for.
MIL-S-22698 - Steel Plate, Carbon, Structural, for Ships.

STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods.

MILITARY

MIL-STD-271 - Nondestructive Testing Requirements for Metals.

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PUBLICATION

NAVAL SEA SYSTEMS COMMAND
NAVSEA 0900-LP-003-9000 - Radiographic Standards for Production and Repair Welds.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 34 - Magnetic Materials, Testing.
A 36 - Structural Steel, Spec for.
A 131 - Structural Steel for Ships.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)
B 4.0 - Standard Methods for Mechanical Testing of Welds.
Z 49.1 - Safety in Welding and Cutting.

(Application for copies should be addressed to the American Welding Society Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

DEPARTMENT OF LABOR
Code of Federal Regulations, Title 29
Part 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Electrodes. Electrodes furnished under this specification shall conform to the requirements of MIL-E-22200 and as specified herein.

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3.2 Covering composition and iron content.

3.2.1 Composition. The chemical composition of the coverings, except as specified in 3.2.2, is optional with the manufacturer. The water content of the coverings shall not exceed the percent-by-weight values specified in table II.

TABLE II. Water content.

Type	Percent by weight maximum
MIL-7018	0.40
MIL-8018-C3	.40
MIL-8018-C3SR	.40
MIL-9018-M	.15
MIL-10018-M	.15
MIL-11018-M	.15

3.2.2 Total iron content. The total iron content of the covering, including any combined iron in addition to metallic iron powder, shall be not less than 15 percent.

3.3 Groove welds.

3.3.1 Mechanical properties. The weld metal properties for the tensile and impact tests required for qualification inspection, comparison inspection, quality conformance inspection or the acquisition document shall be as specified in table III. The applicable stress-relieving treatments shall be as follows:

(a) MIL-7018 and MIL-8018-C3. The coupon shall be heated to 1150°F ± 25°F, held at this temperature for two hours and furnace cooled at 200°F per hour or slower to below 500°F.

(b) MIL-8018-C3SR. The coupon shall be heated to 1150°F ± 25°F, held at this temperature for two hours and furnace cooled at 200°F per hour or slower to below 500°F.

(c) MIL-8018-C3SR. The coupon shall be heated to 1150°F ± 25°F, held at this temperature for 50 hours minimum and furnace cooled at 10°F per hour or slower to 600°F.

(d) MIL-9018-M, MIL-10018-M and MIL-11018-M. The coupon shall be heated to 1025°F ± 25°F, held at this temperature for one hour and furnace cooled at 200°F per hour or slower to below 500°F.

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TABLE III. Mechanical properties.

Type	Ultimate tensile strength		Yield strength[1/]		Elongation[2/] percent in 2 inches		Charpy-V notch impact			
							As-welded		Stress-relieved	
	As-welded	Stress-relieved	As-welded[3/]	Stress-relieved	As-welded	Stress-relieved				
	lb/in^2 minimum	lb/in^2 minimum	lb/in^2	lb/in^2 minimum	Minimum	Minimum	Energy ft. lb. min. avg.	Temperature minus °F	Energy ft. lb. min. avg.	Temperature minus °F
MIL-7018	70,000	-------	58,000 to 70,000	55,000	24	24	20	20	20	20
MIL-8018-C3	80,000	-------	68,000 to 80,000	65,000	24	24	20	20	20	20
MIL-8018-C3SR	-------	80,000	----------	65,000	—	24	—	—	20	20
MIL-9018-M	90,000	-------	78,000 to 90,000	75,000	24	24	20	60	20	60
MIL-10018-M	100,000	-------	88,000 to 100,000	85,000	20	20	20	60	20	60
MIL-11018-M	110,000	-------	98,000 to 110,000	95,000	20	20	35	60	20	60

1/ Yield strength shall be determined by offset method (offset = 0.2 percent).
2/ Percent reduction in area shall be reported for information only.
3/ The upper value of the yield strength may be 5,000 pounds per square inch (lb/in^2) higher than the indicated value for the 3/32 inch size electrode.

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3.4 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table IV.

TABLE IV. Chemical composition (percent)[1/] of deposited weld metal.[2/]

Element	MIL-7018	MIL-8018-C3 MIL-8018-C3SR	MIL-9018-M	MIL-10018-M	MIL-11018-M
Carbon	0.12	0.12	0.10	0.10	0.10
Manganese	.40 - 1.25	.40 - 1.25	.60 - 1.25	.75 - 1.70	1.30 - 1.80
Silicon	.80	.80	.80	.60	0.60
Phosphorus	.030	.030	.030	.030	.030
Sulfur	.030	.030	.030	.030	.030
Chromium	.15	.15	.15	.35	.40
Nickel	.25	.80 - 1.10	1.40 - 1.80	1.40 - 2.10	1.25 - 2.50
Molybdenum	.35	.35	0.35	0.25 - 0.50	0.25 - 0.50
Vanadium	.05	.05	.05	.05	.05

1/ Single values are maximum percentages.
2/ Use of the suffix RC, e.g., MIL-10018-M RC, indicates that the maximum copper content of the deposited weld metal shall be 0.15 percent. The remaining basic compositional requirements remain unchanged for the specific MIL-type.

3.5 Electrode identification. Electrodes shall be identified as specified in 3.5.10 of MIL-E-22200.

3.5.1 Heat and lot identification. When specified (see 6.2), each electrode shall be marked with heat or lot identification as specified in 3.5.10.1 of MIL-E-22200.

3.6 Electrode lengths. The specified lengths of electrodes shall be in accordance with table V, except for type MIL-8018-C3SR which may be qualification tested in longer lengths for the QPL (see 6.2).

TABLE V. Electrode lengths.

Sizes	Electrode lengths end-grip
Inch	Inches
3/32	9, 12 or 14
1/8, 5/32, 3/16	14
7/32, 1/4, 5/16	18

3.7 Alloy identity. The alloy identity provisions of 3.6.4 of MIL-E-22200 shall be applied to each length of rod or wire to be spliced and to both ends of each wire coil drawn to finished size prior to cutting into electrodes. When specified (see 6.2), the core wire of each covered electrode shall meet the alloy identity requirements of 3.6.5 of MIL-E-22200.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-E-22200 and as specified herein.

4.2 Qualification tests. Qualification tests shall be as specified in 4.2 of MIL-E-22200 and table X herein. Types MIL-7018, MIL-8018-C3, MIL-9018-M, MIL-10018-M, and MIL-11018-M shall be tested in both the as-welded and stress-relieved conditions. Type MIL-8018-C3SR shall be tested in each of the stress-relieved conditions specified in 3.3.1(b) and (c). MIL-8018-C3SR shall be tested in the longest length intended to be used (see 3.6).

4.3 Comparison inspection. In addition to the examination and tests required in 4.5 of MIL-E-22200, tests specified in table VI shall be conducted as specified in 4.3 of MIL-E-22200, when required by the Naval Sea Systems Command (NAVSEA).

TABLE VI. Comparison inspection.

Types	Sizes	Tests	Requirements
All All	All All	4.6.8 (MIL-E-22200) Weld test No. 3 (see table X herein)	3.4 and table IV herein 3.3.1 and table III herein

4.4 Quality conformance inspection.

4.4.1 Lot.

4.4.1.1 A lot of electrodes shall be as specified in 4.4.4 and 4.4.6 of MIL-E-22200 for inspection level C for types MIL-7018 and MIL-8018-C3.

4.4.1.2 A lot of electrodes shall be as specified in 4.4.3 and 4.4.6 of MIL-E-22200 for inspection level B for types MIL-9018M, MIL-10018-M, and MIL-11018-M.

4.4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be in accordance with 4.4.7 through 4.4.8.3 of MIL-E-22200 for tests specified in MIL-E-22200 where required and as specified in 4.4.3 for quality conformance weld tests.

4.4.3 Quality conformance inspection.

4.4.3.1 Quality conformance tests shall be conducted on samples as follows:

(a) Types MIL-7018, MIL-8018-C3, and MIL-8018-C3SR. Samples shall be selected as specified in 4.4.2. One sample representing one wet mix of covering mixture in each lot shall be subjected to all the tests required by table VII (i.e., X-ray, mechanical and complete chemical analysis). Samples representing each of the remaining wet mixes of

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covering mixture in a lot shall be tested as specified in table VIII, to assure equivalency as required by 4.4.4.2 of MIL-E-22200.

(b) Types MIL-9018-M, MIL-10018-M, and MIL-11018-M. Samples shall be selected as specified in 4.4.2 from electrodes produced from each wet mix of covering mixture in each lot as defined in 4.4.3 of MIL-E-22200. One sample shall be subjected to all the tests specified in table VII (i.e., X-ray, mechanical and complete chemical analysis). The samples representing each of the remaining wet mixes of covering mixture in each lot shall be subjected to equivalency tests as specified in table VIII and IX.

TABLE VII. Quality conformance inspection.

Types	Tests	Procedures and requirements
All	Test No. 3 and 8a 1/ 2/ 3/ 4/	Table X
All	Table XI	Table XI
All	Chemical analysis 5/	Test No. 3, table X

1/ As-welded impact properties shall be determined only when specifically required by the acquisition document for types MIL-7018, MIL-8018-C3, MIL-9018-M, MIL-10018-M, and MIL-11018-M.
2/ Stress-relieved mechanical and impact properties shall be determined only when specifically required by the acquisition document for types MIL-7018, MIL-8018-C3, MIL-9018-M, MIL-10018-M, and MIL-11018-M.
3/ Type MIL-8018-C3SR shall be tested in the stress-relieved condition only using treatment specified in 3.3.1(b) or (c) as required (see 6.2).
4/ When specified (see 6.2), types MIL-7018, MIL-9018-M, and MIL-11018-M shall have the SR suffix added to the designation, and quality conformance inspected in the stress-relieved condition only, in accordance with 3.3.1(c).
5/ Add a copper analysis for electrode types with suffix RC.

TABLE VIII. Wet mix equivalency tests.

Types	Tests	Procedures
MIL-7018, MIL-8018-C3, and MIL-8018-C3SR MIL-9018-M, MIL-10018-M, MIL-11018-M	1/ Chemical analysis 2/ 3/ 4/ 5/	1/ Table X

1/ See 4.4.4.2 of MIL-E-22200.
2/ Required chemical analysis of the deposited weld metal is for selected elements only as specified in table IX.
3/ Equivalency chemical analysis tests are not required when the lot is as defined in 4.4.3.1(c) of MIL-E-22200.
4/ 4.6.8.1 of MIL-E-22200 may be applied.
5/ Rebaking as specified by footnote 1 to table VII is not required.

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TABLE IX. Wet mix equivalency weld deposit chemical analysis.

	MIL-9018-M	MIL-10018-M	MIL-11018-M
Manganese	X	X	X
Chromium	---	---	---
Nickel	X	X	X
Molybdenum	X	X	X

4.4.3.2 Unsatisfactory test results. Retest shall be in accordance with 4.7 of MIL-E-22200.

4.5 Test procedures.

4.5.1 Weld tests. Table X summarizes the weld tests required under this specification for qualification testing of electrodes and for the inspection as specified in tables VI, VII, VIII, and IX. Medium steel base material in accordance with MIL-S-22698, ASTM A 34, ASTM A 36, and ASTM A 131, grade C plate may be used for all electrode tests.

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TABLE X. Summary of required weld tests.[1/]

Weld test No.	Type	Diameter sizes							Weld joint design procedure	Welding position[2/]	Tests			Chemical	Requirements
		3/32	1/8	5/32	3/16	7/32	1/4	5/16			X-ray	Tensile[4/]	Impact		
3	All	X	X	X	X	X	X	X	Figure 1 and notes	F	3/	X	X	Chemical analysis[5/]	3.7.1 of MIL-E-22200 and tables III and IV herein
8a	All	X ----	X ---	X ----	---- X	---- X	--- X	---- X	Figure 2 and notes	V F	X X[3/]	--- ---	--- ---	---------- ----------	3.7.1 of MIL-E-22200

1/ For qualification testing of the electrodes for the QPL, all electrode types shall be tested using direct current (d.c.), reverse polarity (rp). In addition, MIL-7018 and MIL-8018-C3 electrode types shall be tested using alternating current (a.c.). For comparison inspection, when required, and quality conformance inspection, all electrode types shall be tested using d.c., rp.

2/ V = vertical, F = flat.

3/ When both tests (3 and 8a) are applicable, radiography of test No. 3 will satisfy the requirements for test No. 8a for class 2 and 3 electrodes. If test No. 3 is radiographed, test No. 8a may be eliminated for class 2 and 3 electrodes.

4/ Included ultimate, yield, elongation, reduction in area.

5/ May be determined from milling sample taken from ruptured ends of tensile specimens or drillings. In event of variances with composition requirements or conflicting results by testing activities, analysis shall be verified by procedures specified in 4.6.8 of MIL-E-22200.

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4.5.2 Other tests. Table XI summarizes tests, in additon to those specified in table X required for qualification testing of the electrodes for the QPL and for quality conformance inspection, as applicable.

TABLE XI. Summary of other tests.

Tests	Procedures	Requirements
	MIL-E-22200	
Visual and dimensional	4.6.2.3	3.10 (MIL-E-22200 and 3.6 herein)
Concentricity	4.6.2	3.5.2 (MIL-E-22200)
Dielectric strength[1/]	4.6.3	3.5.4 (MIL-E-22200)
Flaking and cracking of covering[1/]	4.6.4	3.5.5 (MIL-E-22200)
Covering moisture	4.6.9	3.2.1 herein
Total iron content in covering[1/]	4.6.10	3.2.2 herein
Alloy identity	4.6.11	3.6.4 (MIL-E-22200)

[1/] Required for qualification inspection only.

* 4.6 Certification of quality conformance. A certification of quality conformance of the applicable weld tests specified in 4.4.3 and 4.5, containing actual quantitative test and inspection results for quality conformance inspection for each accepted lot, heat, and wet mix, as applicable, included in a particular shipment shall be furnished to the consignee with the shipment. The minimum quality conformance test result data and wet mix equivalence test result data required shall be as shown on figures 3 and 4 and may be submitted in the format of figures 3 and 4.

* 4.6.1 The identification of melts, heats or lots; the general requirements for test specimens; the rejection of nonconforming lots or specimens; and the retest and resubmittal of nonconforming lots shall be in accordance with FED-STD-151.

4.7 Alloy identity certification. When specified (see 6.2), the inspector shall affix his stamp and date on the container, containing electrodes conforming to 3.6.5 of MIL-E-22200. The letters "AT" shall be affixed to the container adjacent to the date to indicate "Alloy Tested".

4.8 Packaging inspection. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

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5.1.1 Level A. All electrodes shall be packaged in class 1 unit containers in accordance with MIL-W-10430, except when specified (see 6.2), 12-, 14-, and 18-inch electrodes shall be supplied in class 1b (50-pound) unit containers.

5.1.1.1 Special instructions. When electrodes are packaged in class 1a unit containers, in accordance with MIL-W-10430, fiberboard or other cushioning material shall be inserted in each end of the unit container to restrict the end-wise movement of the electrodes. The cushioning materials shall be dry and in equilibrium with the electrode covering. Class 1a containers shall be of a snug fitting design with the container length not to exceed the rod length by approximately 1/2-inch.

5.1.2 Level C. Packaging shall afford protection against deterioration and physical damage during shipment from the supply source to the using activity. The packaging applied shall insure conformance to the stability requirements specified in 3.5.8 of MIL-E-22200.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A and B. Electrodes, packaged as specified (see 6.2), shall be packed level A or B in accordance with MIL-W-10430. Palletization, when required, shall be in accordance with MIL-W-10430.

5.2.2 Level C. Electrodes, packaged as specified (see 6.2), shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing or method of shipment shall comply with Uniform Freight Classification Rules and Regulations or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required herein or by the contract or order (see 6.2), interior packages and exterior shipping containers and palletized loads shall be marked in accordance with MIL-W-10430.

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5.3.1 Special marking.

* 5.3.1.1 Interior packages and shipping containers. In addition to the lot identification required by 4.4.6 of MIL-E-22200, identification on interior packages shall include MIL type, size, specification number, manufacturer's or distributor's name, date of manufacture (month/year), and manufacturer's or distributor's brand or type designation. In addition, all unit packages shall be marked or labeled with the following precautionary information, as a minimum and prominently displayed:

WARNING: Protect yourself and others. Read and understand this label.

FUMES AND GASES can be dangerous to your health. ARC RAYS can injure eyes and burn skin. ELECTRIC SHOCK can kill.

- Read and understand the manufacturer's instructions and your employer's safety practices.
- Keep your head out of the fumes.
- Use enough ventilation, exhaust at the arc, or both to keep fumes and gases from your breathing zone, and the general area.
- Wear correct eye, ear and body protection.
- Do not touch live electrical parts.
- See American National Standard Z49.1 "Safety in Welding and Cutting" published by the American Welding Society, 2501 N.W. 7th St., Miami, FL, 33125; OSHA Safety and Health Standards, 29 CRF 1910, available from U.S. Dept. of Labor, Washington, DC 20210.

DO NOT REMOVE THIS LABEL

5.3.1.2 All containers shall include the following marking:

"ASBESTOS FREE"

5.3.1.3 Class 1b containers. Each class 1b container on the container end adjacent to the electrode grip end shall be marked "OPEN THIS END".

6. NOTES

6.1 Intended use. This specification is intended to cover several low-hydrogen types of arc-welding electrodes of various alloys and strength levels, capable of depositing weld metal, that in the as-welded or stress-relieved conditions meet mechanical-property requirements given herein. It will be noted that, although these electrodes may be used for some applications in the as-welded condition, they are capable of stress relief as required. Properties may be affected by long time stress-relieving and slow cooling from stress-relieving temperature. Properties should be determined to be satisfactory after a simulated production postweld heat treatment by the user when requird by the fabrication document. The designation, low-hydrogen, is applied to electrodes having coverings that are substantially free of any compound that may liberate hydrogen in the

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arc. Low alloy steel electrodes that produce low-hydrogen in the arc have been found suitable for welding many steels that formerly gave serious difficulty through excessive porosity and underbead cracking, unless welded with high-alloy electrodes.

6.1.1 Type MIL-7018. This type is intended to cover medium steel electrodes used in welding mild, medium or high-tensile steels such as grade HT of MIL-S-16113 and MIL-S-22698.

6.1.2 Type MIL-8010-C3. This type is intended to cover low-alloy electrodes capable of depositing weld metal of higher strength than type MIL-7018.

6.1.3 Type MIL-8018-C3SR. This type is intended to cover a special MIL-type electrode formulated to meet the long time (50-hours) stress-relieved mechanical-properties requirements only.

6.1.4 Type MIL-9018-M. This type is intended to cover low-alloy electrodes suitable for welding steels having properties similar to those of the weld, and for welding attachments to HY-80 steel, as may be permitted. This electrode is intended to be used with the d.c., rp electrical characteristics only.

6.1.5 Type MIL-10018-M. This type is intended to cover low-alloy electrodes for welding low-alloy, high-yield, notch-tough steel such as grade HY-80 of MIL-S-16216. This electrode is intended to be used with d.c., rp electrical characteristics only.

6.1.6 Type MIL-11018-M. This type is intended to cover low-alloy electrodes for welding groove butt joints in HY-80 and T-1 steels. This electrode is intended to be used with d.c., rp electrical characteristics only.

6.1.7 All types with suffix RC. This type is intended to cover electrodes for welding pressure and strength welds which will be subject to irradiation.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class and size required (see 1.2).
(c) Whether the copper content is restricted as designated by the suffix RC to the MIL-type (see 3.4).
(d) Whether electrode identification shall include heat or lot identification (see 3.5.1).
(e) For MIL-8018-C3SR, the length required (see 3.6).
(f) For 3/32-inch size electrode, the length required (see table V).
(g) Whether quality conformance impact tests are required (see table VII, footnote 2).
(h) Whether stress-relieved quality conformance inspection are required (see table VII, footnote 2).
(i) For MIL-8018-C3SR, the stress-relieved condition required for quality conformance inspection (see table VII, footnote 3).

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(j) Whether types MIL-7018, MIL-9018-M, MIL-10018-M, or MIL-11018-M are to be quality conformance inspected in the stress-relieved condition for the 50-hour holding time (see table VII, footnote 4).
(k) Levels of packaging and packing required (see 5.1 and 5.2).
(l) When 12-, 14- and 18-inch electrodes are required in class 1b containers (see 5.1.1).
(m) Special marking required (see 5.3).
(n) Whether the alloy identity test and certification is required for the core wire of each finished electrode (see 3.7 and 4.7).

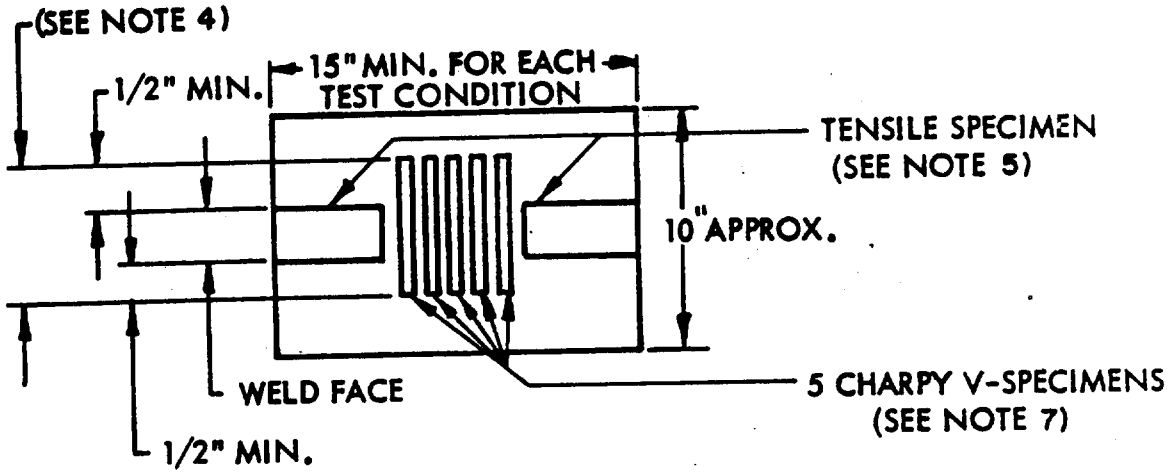
6.2.1 Electrodes should be ordered by the pound.

6.3 Inspection after delivery. Post delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of ths document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

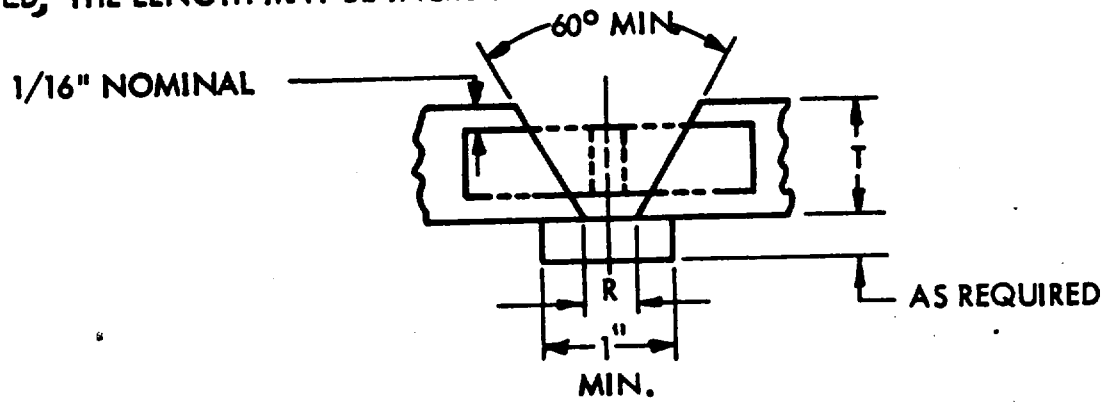
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LAYOUT OF TEST PLATE

NOTE: LENGTH MAY BE DECREASED ACCORDINGLY WHERE IMPACTS OR STRESS RELIEVED SAMPLES ARE NOT REQUIRED. WHERE IMPACTS IN THE AS-WELDED AND STRESS-RELIEVED CONDITION BOTH ARE REQUIRED, THE LENGTH MAY BE INCREASED ACCORDINGLY



GROOVE PREPARATION OF TEST PLATE

SHOWING ORIENTATION OF CHARPY-V TEST SPECIMEN

SH 8985

FIGURE 1. Unrestricted welded joint for mechanical and impact tests (test no. 3).

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Notes to figure 1:

1. Base metal shall be as specified in 4.5.1.
2. Welds shall be deposited in layers using a 2-1/2 core wire diameter maximum weave build-up sequence. Record and report as specified in table XII. Layers should be approximately 1/8-inch thick, and shall be deposited in such a manner that the final layers include a reinforcement of standard proportions (1/32-inch to 1/8-inch). Welding currents shall be in accordance with the manufacturer's recommendations and the actual values used shall be recorded and reported. Each layer shall be started at the finishing end of the preceding layer.

TABLE XII. Groove weld preparation.

Electrode size	Minimum thickness (T)	Maximum root opening (R)	Number of layers 1/	
			Minimum	Maximum
(Inch)	(Inches)	(Inch)		
3/32	1/2	1/2	1/	1/
1/8	1/2	1/4	1/	1/
5/32	3/4	1/2	7	9
3/16	3/4	1/2	7	9
7/32	3/4	1/2	7	8
1/4	1	1/2	9	11
5/16	1-1/4	1/2	11	12

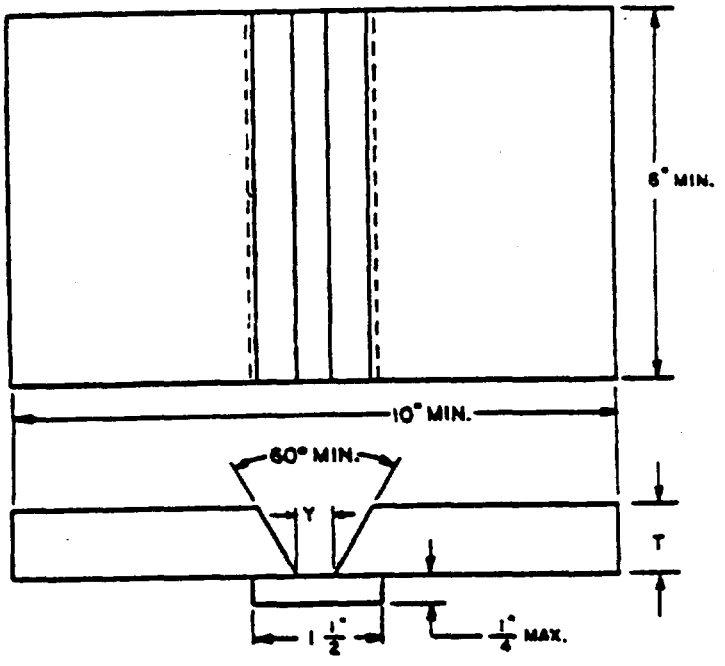
1/ Pass and layer sequence shall be reported.

3. After the joint has been tacked, the assembly shall be heated to 225°F ± 25°F prior to welding. After completion of each pass the assembly shall be allowed to cool in still air at room temperature and the subsequent pass shall be started when the temperature of the plate has dropped 225°F ± 25°F. If it is necessary to interrupt this procedure, the assembly shall be preheated to 225°F ± 25°F.
4. The weld shall be cut as shown resulting in two tensile coupons suitable for 0.505-inch diameter specimens (or 0.252-inch diameter specimens when testing 3/32-inch and 1/8-inch size electrodes) and a center coupon of size sufficient for the required Charpy V-notch specimens. No base metal shall be removed within 1/2-inch of the edges of the face of the weld by flame cutting. Only sawing or machining shall be used for specimen finishing.
5. One tensile coupon, where required, shall be stress-relieved in accordance with 3.3.1 herein.
6. Both the stress-relieved and as-welded coupons, when required, shall be machined into 0.505-inch tensile specimens (or 0.252-inch diameter specimens when testing 3/32-inch and 1/8-inch sie electrodes). For details see AWS B4.0.

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7. The center section of the welded joint shall be used for the preparation of Charpy V-notch impact specimens. The Charpy V-notch specimens shall be machined in accordance with AWS B4.0 details shown on figure 1. Prior to machining specimens, where required, the coupon shall be stress-relieved in accordance with 3.3.1 herein.
8. Impact properties for the five specimens for each test condition shall be obtained at test temperatures specified in table III herein, plus or minus 3°F. The extreme lowest impact value thus obtained together with the extreme highest value may be disregarded for this test. Two of the three remaining values shall be greater than that specified in table III herein. One of the three remaining values may be 5 foot/pounds less than that specified in table III herein. The computed average value of the three remaining values shall be greater than that specified in table III herein.

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SH 11730

Electrode size	3/32 (Inch)	1/8 (Inch)	5/32 (Inch)	3/16-5/16 (Inch)
Thickness (T)1/	1/4	3/8	1/2	1/2
Root opening (Y)2/	1/4	5/16	3/8	1/2
Weld bead width (max)	3/8	1/2	5/8	3/4

1/ Minimum.
2/ Tolerance ± 1/16-inch.

FIGURE 2. Groove weld useability test (test 8a).

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Notes to figure 2:

1. Welding shall be accomplished with the test plate in the vertical or flat position as specified in table X. Welding current shall be in accordance with the manufacturer's recommendations and the actual values shall be recorded and reported. Unless otherwise specified by the acquisition document, test shall be done with d.c. only.
2. Base metal shall be in accordance with 4.5.1 herein.
3. The edge preparation if flame cut shall be ground to remove slag and surface irregularities.
4. Welds shall be deposited in approximately 1/8-inch thick layers. Weld bead widths shall be as shown above. Each bead shall contain a start in the area to be evaluated.
5. The reinforcement shall be 3/16-inch maximum.
6. Preheat and interpass temperature shall be 225°F $\pm$ 25°F.
7. After completion of the weld, the weld reinforcement and backing strips shall be removed flush with the base plate on both surfaces and the assembly shall be X-rayed in accordance with MIL-STD-271.
8. The radiograph shall meet the X-ray standards of NAVSEA 0900-LP-003-9000 as specified in 3.7.1 of MIL-E-22200.

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CERTIFICATION OF QUALITY CONFORMANCE TESTS

Manufacturer or Distributor _____

Address _____

Date _____

Specification MIL-_____

Type MIL-_____

Diameter & Length _____

Inspection Level _____

Lot No. _____

Customer's Name _____

Customer's Order No. _____

Core Wire Heat No. _____

Lot Identification MIL-E-22200 Para. _____

Wet Batch No. _____ _____

Chemical Analysis (Complete)

Carbon _____

Manganese _____

Silicon _____

Phosphorus _____

Sulfur _____

Chromium _____

Nickel _____

Molybdenum _____

Vanadium _____

Copper _____

Mechanical Tests

	As-welded	Stress-relieved
Yield Strength (0.2% offset method)		
Tensile Strength	_____	_____
Elongation (%)	_____	_____
Reduction of Area (%)	_____	_____
	_____	_____
CHARPY IMPACTS		
1.	_____	_____
2.	_____	_____
3.	_____	_____
4.	_____	_____
5.	_____	_____

Alloy identity tests:

(a) Core wire _____

(b) Finished electrodes _____ (If required)

FIGURE 3. Certificate of quality conformance.

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Chemistry was taken from: ☐ Chem Pad ☐ Groove Weld

X-ray Results_____

Concentricity (%)_____

Covering Moisture_____

Grinding during 8a test plate preparation (see note 3)_____

Groove Weld Test

Test No. 3 8a Chem Pad

Amperage ____ _____ _____

Operator Error (Layer Nos.)_____

We hereby certify that the above material has been tested in accordance with the listed specification and is in conformance with all requirements.

FIGURE 3. Certificate of quality conformance. - Continued

Notes to figure 3:

1. Items not applicable to the type electrode involved or exempted by the acquisition document shall be marked N/A.
2. This form is applicable to the quality conformance tests required for each lot. Figure 4 forms shall be employed to report and certify the wet mix equivalency tests where applicable for each wet mix of covering mixture in the lot. One copy of the figure 4 form shall be employed for certifying equivalency of each wet mix in a lot where applicable. One copy of the quality conformance certificate (figure 3) for the lot shall be forwarded with copies of the batch equivalency certificates for each additional mix from that lot that is included in the shipment.
3. See paragraphs 3.5.7 and 3.9 of MIL-E-22200.
4. Copper content is required for MIL types with RC suffix.

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CERTIFICATION OF WET MIX EQUIVALENCY TESTS

Manufacturer or Distributor_____

Address_____

Date_____

Specification MIL-_____

Type MIL-_____

Diameter Size_____

Lot No._____

Customer's Name_____

Customer's Order No._____

Core Wire Heat No._____

Wet Batch No._____

Sampling
Chemical Analysis

Manganese_____

Silicon_____

Chromium_____

Nickel_____

Molybdenum_____

Copper_____

X-Ray Results_____

Chemistry was taken from:

☐ Chem Pad ☐ Groove Weld

Test No._____

Chem Pad 8a

Amperage _____ _____ _____

We hereby certify that the above material has been tested in accordance with the listed specification and in conformance with all requirements.

FIGURE 4. Certificate of wet mix equivalency tests.

Notes to figure 4:

1. Items not applicable to the type electrode involved or exempted by the acquistion document shall be marked N/A.
2. One copy of the figure 4 form shall be employed for certifying equivalency of each wet mix in a lot where applicable. One copy of the quality conformance certificate (figure 3) for the lot shall be forwarded with copies of the mix equivalency certificates for each additional mix from that lot that is included in the shipment.
3. Copper content is required for MIL-types with RC suffix.

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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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